ZOO501T-Devekopmental Biology

Assignment 3

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Note:Model should be manually made and then Captured image of that will have to be uploded in LMS Inline Interface Assignment Submission.

This guide provides step-by-step instructions for crafting an educational model showcasing the stages of sea urchin gastrulation.

Materials

- Multicolored clay or playdough
- Toothpicks or small sticks for labels
- Marker or pen
- Small balls (to represent blastomeres)
- Scissors and cutting tools (optional)

Construction Process

1. Fertilized Egg Stage

- Form a small clay ball in white or a light color

2. Cleavage Stage

- Divide the initial model into progressively smaller sections
- Use different colors for each division (2-cell, 4-cell, 8-cell stages)

3. Blastula Stage

- Create a hollow clay sphere
- Attach small clay balls around the surface to represent blastomeres
- Maintain color consistency with the fertilized egg stage

4. Gastrula Stage

- Form an indentation (blastopore) in the hollow sphere
- Extend the indentation inward to show invagination
- Shape the interior to form the archenteron (primitive gut)
- Use a contrasting color for the invaginated area

Labeling and Display

1. Create paper flags on toothpicks for each stage
2. Label each model with its corresponding stage name
3. Arrange models sequentially on a display board or flat surface
4. Ensure clear visibility and labeling for each model

Helpful Tips

- Reference images of sea urchin gastrulation for accuracy
- Focus on maintaining proper symmetry and proportions
- Emphasize key structural changes between stages

This hands-on project aids in visualizing and comprehending the intricate process of sea urchin gastrulation, enhancing both learning and retention of the subject matter.



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